Chapter

4 Expectation

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4.1 The Expectation of a Random Variable

The distribution of a random variable X contains all of the probabilistic information about X. The entire distribution of X, however, is usually too cumbersome for presenting this information. Summaries of the distribution, such as the average value, or expected value, can be useful for giving people an idea of where we expect X to be without trying to describe the entire distribution. The expected value also plays an important role in the approximation methods that arise in Chapter 6.

Expectation for a Discrete Distribution

Example 4.1.1

Fair Price for a Stock. An investor is considering whether or not to invest \$18 per share in a stock for one year. The value of the stock after one year, in dollars, will be $18 + X$, where X is the amount by which the price changes over the year. At present X is unknown, and the investor would like to compute an "average value" for X in order to compare the return she expects from the investment to what she would get by putting the \$18 in the bank at 5% interest. ◀

The idea of finding an average value as in Example 4.1.1 arises in many applications that involve a random variable. One popular choice is what we call the *mean* or *expected value* or *expectation*.

The intuitive idea of the mean of a random variable is that it is the weighted average of the possible values of the random variable with the weights equal to the probabilities.

Example 4.1.2

Stock Price Change. Suppose that the change in price of the stock in Example 4.1.1 is a random variable X that can assume only the four different values -2, 0, 1, and 4, and that $\Pr(X = -2) = 0.1$, $\Pr(X = 0) = 0.4$, $\Pr(X = 1) = 0.3$, and $\Pr(X = 4) = 0.2$. Then the weighted avarage of these values is

$$-2(0.1) + 0(0.4) + 1(0.3) + 4(0.2) = 0.9.$$

The investor now compares this with the interest that would be earned on \$18 at 5% for one year, which is $18 \times 0.05 = 0.9$ dollars. From this point of view, the price of \$18 seems fair. ◀

The calculation in Example 4.1.2 generalizes easily to every random variable that assumes only finitely many values. Possible problems arise with random variables that assume more than finitely many values, especially when the collection of possible values is unbounded.

Definition 4.1.1 Mean of Bounded Discrete Random Variable. Let X be a bounded discrete random variable whose p.f. is f. The *expectation of* X, denoted by $E(X)$, is a number defined as follows:

$$E(X) = \sum_{\text{All } x} xf(x). \tag{4.1.1}$$

The expectation of X is also referred to as the *mean of* X or the *expected value of* X.

In Example 4.1.2, $E(X) = 0.9$. Notice that 0.9 is not one of the possible values of X in that example. This is typically the case with discrete random variables.

Example 4.1.3 Bernoulli Random Variable. Let X have the Bernoulli distribution with parameter p, that is, assume that X takes only the two values 0 and 1 with $\Pr(X = 1) = p$. Then the mean of X is

$$E(X) = 0 \times (1 - p) + 1 \times p = p. \quad \blacktriangleleft$$

If X is unbounded, it might still be possible to define $E(X)$ as the weighted average of its possible values. However, some care is needed.

Definition 4.1.2 Mean of General Discrete Random Variable. Let X be a discrete random variable whose p.f. is f. Suppose that at least one of the following sums is finite:

$$\sum_{\text{Positive } x} xf(x), \qquad \sum_{\text{Negative } x} xf(x). \tag{4.1.2}$$

Then the *mean*, *expectation*, or *expected value* of X is said to *exist* and is defined to be

$$E(X) = \sum_{\text{All } x} xf(x). \tag{4.1.3}$$

If both of the sums in (4.1.2) are infinite, then $E(X)$ *does not exist*.

The reason that the expectation fails to exist if both of the sums in (4.1.2) are infinite is that, in such cases, the sum in (4.1.3) is not well-defined. It is known from calculus that the sum of an infinite series whose positive and negative terms both add to infinity either fails to converge or can be made to converge to many different values by rearranging the terms in different orders. We don't want the meaning of expected value to depend on arbitrary choices about what order to add numbers. If only one of two sums in (4.1.3) is infiinte, then the expected value is also infinite with the same sign as that of the sum that is infinite. If both sums are finite, then the sum in (4.1.3) converges and doesn't depend on the order in which the terms are added.

Example 4.1.4 The Mean of X Does Not Exist. Let X be a random variable whose p.f. is

$$f(x) = \begin{cases} \dfrac{1}{2|x|(|x|+1)} & \text{if } x = \pm 1, \pm 2, \pm 3, \ldots, \\ 0 & \text{otherwise.} \end{cases}$$

It can be verified that this function satisfies the conditions required to be a p.f. The two sums in (4.1.2) are

$$\sum_{x=-1}^{-\infty} x\frac{1}{2|x|(|x|+1)} = -\infty \quad \text{and} \quad \sum_{x=1}^{\infty} x\frac{1}{2x(x+1)} = \infty;$$

hence, $E(X)$ does not exist. ◀

Example 4.1.5 An Infinite Mean. Let X be a random variable whose p.f. is

$$f(x) = \begin{cases} \dfrac{1}{x(x+1)} & \text{if } x = 1, 2, 3, \ldots, \\ 0 & \text{otherwise.} \end{cases}$$

The sum over negative values in Eq. (4.1.2) is 0, so the mean of X exists and is

$$E(X) = \sum_{x=1}^{\infty} x\frac{1}{x(x+1)} = \infty.$$

We say that the mean of X is *infinite* in this case. ◀

Note: The Expectation of X Depends Only on the Distribution of X. Although $E(X)$ is called the expectation of X, it depends only on the distribution of X. Every two random variables that have the same distribution will have the same expectation even if they have nothing to do with each other. For this reason, we shall often refer to the expectation of a distribution even if we do not have in mind a random variable with that distribution.

Expectation for a Continuous Distribution

The idea of computing a weighted average of the possible values can be generalized to continuous random variables by using integrals instead of sums. The distinction between bounded and unbounded random variables arises in this case for the same reasons.

Definition 4.1.3 Mean of Bounded Continuous Random Variable. Let X be a bounded continuous random variable whose p.d.f. is f. The *expectation of* X, denoted $E(X)$, is defined as follows:

$$E(X) = \int_{-\infty}^{\infty} xf(x)\,dx. \tag{4.1.4}$$

Once again, the expectation is also called the *mean* or the *expected value*.

Example 4.1.6 Expected Failure Time. An appliance has a maximum lifetime of one year. The time X until it fails is a random variable with a continuous distribution having p.d.f.

$$f(x) = \begin{cases} 2x & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$E(X) = \int_0^1 x(2x)\,dx = \int_0^1 2x^2\,dx = \frac{2}{3}.$$

We can also say that the expectation of the distribution with p.d.f. f is 2/3. ◀

For general continuous random variables, we modify Definition 4.1.2.

Definition 4.1.4 Mean of General Continuous Random Variable. Let X be a continuous random variable whose p.d.f. is f. Suppose that at least one of the following integrals is finite:

$$\int_0^\infty xf(x)dx, \quad \int_{-\infty}^0 xf(x)dx. \tag{4.1.5}$$

Then the *mean*, *expectation*, or *expected value* of X is said to *exist* and is defined to be

$$E(X) = \int_{-\infty}^\infty xf(x)dx. \tag{4.1.6}$$

If both of the integrals in (4.1.5) are infinite, then $E(X)$ *does not exist*.

Example 4.1.7 Failure after Warranty. A product has a warranty of one year. Let X be the time at which the product fails. Suppose that X has a continuous distribution with the p.d.f.

$$f(x) = \begin{cases} 0 & \text{for } x < 1, \\ \frac{2}{x^3} & \text{for } x \geq 1. \end{cases}$$

The expected time to failure is then

$$E(X) = \int_1^\infty x\frac{2}{x^3}dx = \int_1^\infty \frac{2}{x^2}dx = 2. \quad \blacktriangleleft$$

Example 4.1.8 A Mean That Does Not Exist. Suppose that a random variable X has a continuous distribution for which the p.d.f. is as follows:

$$f(x) = \frac{1}{\pi(1+x^2)} \quad \text{for } -\infty < x < \infty. \tag{4.1.7}$$

This distribution is called the *Cauchy distribution*. We can verify the fact that $\int_{-\infty}^\infty f(x)\,dx = 1$ by using the following standard result from elementary calculus:

$$\frac{d}{dx}\tan^{-1}x = \frac{1}{1+x^2} \quad \text{for } -\infty < x < \infty.$$

The two integrals in (4.1.5) are

$$\int_0^\infty \frac{x}{\pi(1+x^2)}dx = \infty \quad \text{and} \quad \int_{-\infty}^0 \frac{x}{\pi(1+x^2)}dx = -\infty;$$

hence, the mean of X does not exist. $\blacktriangleleft$

Interpretation of the Expectation

Relation of the Mean to the Center of Gravity The expectation of a random variable or, equivalently, the mean of its distribution can be regarded as being the center of gravity of that distribution. To illustrate this concept, consider, for example, the p.f. sketched in Fig. 4.1. The x-axis may be regarded as a long weightless rod to which weights are attached. If a weight equal to $f(x_j)$ is attached to this rod at each point x_j, then the rod will be balanced if it is supported at the point $E(X)$.

Now consider the p.d.f. sketched in Fig. 4.2. In this case, the x-axis may be regarded as a long rod over which the mass varies continuously. If the density of

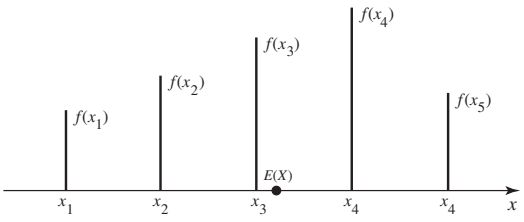


Figure 4.1 The mean of a discrete distribution.

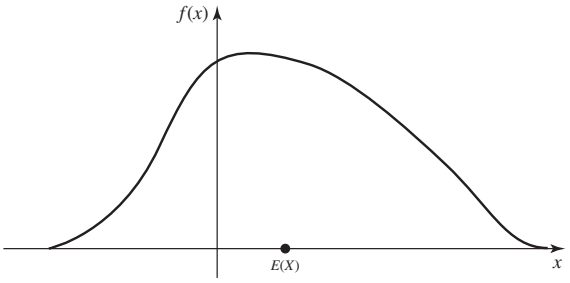


Figure 4.2 The mean of a continuous distribution.

the rod at each point x is equal to $f(x)$, then the center of gravity of the rod will be located at the point $E(X)$, and the rod will be balanced if it is supported at that point.

It can be seen from this discussion that the mean of a distribution can be affected greatly by even a very small change in the amount of probability that is assigned to a large value of x. For example, the mean of the distribution represented by the p.f. in Fig. 4.1 can be moved to any specified point on the x-axis, no matter how far from the origin that point may be, by removing an arbitrarily small but positive amount of probability from one of the points x_j and adding this amount of probability at a point far enough from the origin.

Suppose now that the p.f. or p.d.f. f of some distribution is symmetric with respect to a given point x_0 on the x-axis. In other words, suppose that $f(x_0 + \delta) = f(x_0 - \delta)$ for all values of δ. Also assume that the mean $E(X)$ of this distribution exists. In accordance with the interpretation that the mean is at the center of gravity, it follows that $E(X)$ must be equal to x_0, which is the point of symmetry. The following example emphasizes the fact that it is necessary to make certain that the mean $E(X)$ exists before it can be concluded that $E(X) = x_0$.

Example 4.1.9

The Cauchy Distribution. Consider again the p.d.f. specified by Eq. (4.1.7), which is sketched in Fig. 4.3. This p.d.f. is symmetric with respect to the point $x = 0$. Therefore, if the mean of the Cauchy distribution existed, its value would have to be 0. However, we saw in Example 4.1.8 that the mean of X does not exist.

The reason for the nonexistence of the mean of the Cauchy distribution is as follows: When the curve $y = f(x)$ is sketched as in Fig. 4.3, its tails approach the x-axis rapidly enough to permit the total area under the curve to be equal to 1. On the other hand, if each value of $f(x)$ is multiplied by x and the curve $y = xf(x)$ is sketched, as in Fig. 4.4, the tails of this curve approach the x-axis so slowly that the total area between the x-axis and each part of the curve is infinite. ◀

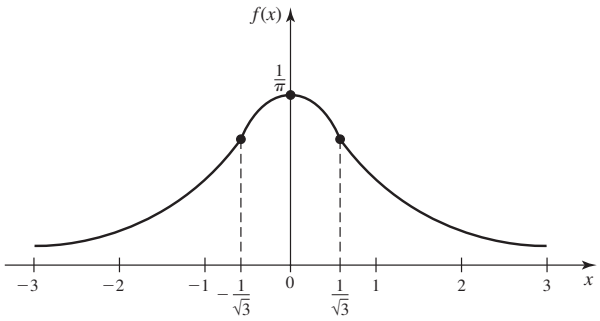


Figure 4.3 The p.d.f. of a Cauchy distribution.

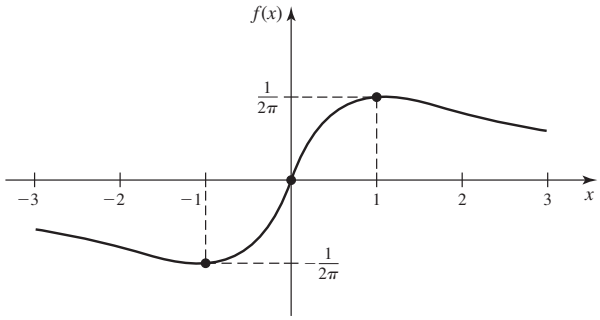


Figure 4.4 The curve $y = xf(x)$ for the Cauchy distribution.

The Expectation of a Function

Example 4.1.10

Failure Rate and Time to Failure. Suppose that appliances manufactured by a particular company fail at a rate of X per year, where X is currently unknown and hence is a random variable. If we are interested in predicting how long such an appliance will last before failure, we might use the mean of $1/X$. How can we calculate the mean of $Y = 1/X$? ◀

Functions of a Single Random Variable If X is a random variable for which the p.d.f. is f, then the expectation of each real-valued function $r(X)$ can be found by applying the definition of expectation to the distribution of $r(X)$ as follows: Let $Y = r(X)$, determine the probability distribution of Y, and then determine $E(Y)$ by applying either Eq. (4.1.1) or Eq. (4.1.4). For example, suppose that Y has a continuous distribution with the p.d.f. g. Then

$$E[r(X)] = E(Y) = \int_{-\infty}^{\infty} yg(y)\,dy, \tag{4.1.8}$$

if the expectation exists.

Example 4.1.11

Failure Rate and Time to Failure. In Example 4.1.10, suppose that the p.d.f. of X is

$$f(x) = \begin{cases} 3x^2 & \text{if } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $r(x) = 1/x$. Using the methods of Sec. 3.8, we can find the p.d.f. of $Y = r(X)$ as

$$g(y) = \begin{cases} 3y^{-4} & \text{if } y > 1, \\ 0 & \text{otherwise.} \end{cases}$$

The mean of Y is then

$$E(Y) = \int_0^\infty y 3y^{-4} dy = \frac{3}{2}. \quad \blacktriangleleft$$

Although the method of Example 4.1.11 can be used to find the mean of a continuous random variable, it is not actually necessary to determine the p.d.f. of $r(X)$ in order to calculate the expectation $E[r(X)]$. In fact, it can be shown that the value of $E[r(X)]$ can always be calculated directly using the following result.

Theorem 4.1.1 Law of the Unconscious Statistician. Let X be a random variable, and let r be a real-valued function of a real variable. If X has a continuous distribution, then

$$E[r(X)] = \int_{-\infty}^{\infty} r(x) f(x)\, dx, \tag{4.1.9}$$

if the mean exists. If X has a discrete distribution, then

$$E[r(X)] = \sum_{\text{All } x} r(x) f(x), \tag{4.1.10}$$

if the mean exists.

Proof A general proof will not be given here. However, we shall provide a proof for two special cases. First, suppose that the distribution of X is discrete. Then the distribution of Y must also be discrete. Let g be the p.f. of Y. For this case,

$$\begin{aligned} \sum_y y g(y) &= \sum_y y \Pr[r(X) = y] \\ &= \sum_y y \sum_{x: r(x) = y} f(x) \\ &= \sum_y \sum_{x: r(x) = y} r(x) f(x) = \sum_x r(x) f(x). \end{aligned}$$

Hence, Eq. (4.1.10) yields the same value as one would obtain from Definition 4.1.1 applied to Y.

Second, suppose that the distribution of X is continuous. Suppose also, as in Sec. 3.8, that $r(x)$ is either strictly increasing or strictly decreasing with differentiable inverse $s(y)$. Then, if we change variables in Eq. (4.1.9) from x to $y = r(x)$,

$$\int_{-\infty}^{\infty} r(x) f(x)\, dx = \int_{-\infty}^{\infty} y f[s(y)] \left| \frac{ds(y)}{dy} \right| dy.$$

It now follows from Eq. (3.8.3) that the right side of this equation is equal to

$$\int_{-\infty}^{\infty} y g(y)\, dy.$$

Hence, Eqs. (4.1.8) and (4.1.9) yield the same value. ■

Theorem 4.1.1 is called the law of the unconscious statistician because many people treat Eqs. (4.1.9) and (4.1.10) as the definition of $E[r(X)]$ and forget that the definition of the mean of $Y = r(X)$ is given in Definitions 4.1.2 and 4.1.4.

Example 4.1.12 Failure Rate and Time to Failure. In Example 4.1.11, we can apply Theorem 4.1.1 to find

$$E(Y) = \int_0^1 \frac{1}{x} 3x^2 dx = \frac{3}{2},$$

the same result we got in Example 4.1.11. ◀

Example 4.1.13 Determining the Expectation of $X^{1/2}$. Suppose that the p.d.f. of X is as given in Example 4.1.6 and that $Y = X^{1/2}$. Then, by Eq. (4.1.9),

$$E(Y) = \int_0^1 x^{1/2}(2x)\, dx = 2\int_0^1 x^{3/2}\, dx = \frac{4}{5}.$$ ◀

Note: In General, $E[g(X)] \neq g(E(X))$. In Example 4.1.13, the mean of $X^{1/2}$ is 4/5. The mean of X was computed in Example 4.1.6 as 2/3. Note that $4/5 \neq (2/3)^{1/2}$. In fact, unless g is a linear function, it is generally the case that $E[g(X)] \neq g(E(X))$. A linear function g does satisfy $E[g(X)] = g(E(X))$, as we shall see in Theorem 4.2.1.

Example 4.1.14 Option Pricing. Suppose that common stock in the up-and-coming company A is currently priced at \$200 per share. As an incentive to get you to work for company A, you might be offered an option to buy a certain number of shares of the stock, one year from now, at a price of \$200. This could be quite valuable if you believed that the stock was very likely to rise in price over the next year. For simplicity, suppose that the price X of the stock one year from now is a discrete random variable that can take only two values (in dollars): 260 and 180. Let p be the probability that $X = 260$. You want to calculate the value of these stock options, either because you contemplate the possibility of selling them or because you want to compare Company A's offer to what other companies are offering. Let Y be the value of the option for one share when it expires in one year. Since nobody would pay \$200 for the stock if the price X is less than \$200, the value of the stock option is 0 if $X = 180$. If $X = 260$, one could buy the stock for \$200 per share and then immediately sell it for \$260. This brings in a profit of \$60 per share. (For simplicity, we shall ignore dividends and the transaction costs of buying and selling stocks.) Then $Y = h(X)$ where

$$h(x) = \begin{cases} 0 & \text{if } x = 180, \\ 60 & \text{if } x = 260. \end{cases}$$

Assume that an investor could earn 4% risk-free on any money invested for this same year. (Assume that the 4% includes any compounding.) If no other investment options were available, a fair cost of the option would then be what is called the *present value* of $E(Y)$ in one year. This equals the value c such that $E(Y) = 1.04c$. That is, the expected value of the option equals the amount of money the investor would have after one year without buying the option. We can find $E(Y)$ easily:

$$E(Y) = 0 \times (1 - p) + 60 \times p = 60p.$$

So, the fair price of an option to buy one share would be $c = 60p/1.04 = 57.69p$.

How should one determine the probability p? There is a standard method used in the finance industry for choosing p in this example. That method is to assume that

the present value of the mean of X (the stock price in one year) is equal to the current value of the stock price. That is, assume that the expected value of buying one share of stock and waiting one year to sell is the same as the result of investing the current cost of the stock risk-free for one year (multiplying by 1.04 in this example). In our example, this means $E(X) = 200 \times 1.04$. Since $E(X) = 260p + 180(1 - p)$, we set

$$200 \times 1.04 = 260p + 180(1 - p),$$

and obtain $p = 0.35$. The resulting price of an option to buy one share for \$200 in one year would be $\$57.69 \times 0.35 = \$20.19$. This price is called the *risk-neutral price of the option*. One can prove (see Exercise 14 in this section) that any price other than \$20.19 for the option would lead to unpleasant consequences in the market. ◀

Functions of Several Random Variables

Example 4.1.15 **The Expectation of a Function of Two Variables.** Let X and Y have a joint p.d.f., and suppose that we want the mean of $X^2 + Y^2$. The most straightforward but most difficult way to do this would be to use the methods of Sec. 3.9 to find the distribution of $Z = X^2 + Y^2$ and then apply the definition of mean to Z. ◀

There is a version of Theorem 4.1.1 for functions of more than one random variable. Its proof is not given here.

Theorem 4.1.2 **Law of the Unconscious Statistician.** Suppose that $X_1, \ldots, X_n$ are random variables with the joint p.d.f. $f(x_1, \ldots, x_n)$. Let r be a real-valued function of n real variables, and suppose that $Y = r(X_1, \ldots, X_n)$. Then $E(Y)$ can be determined directly from the relation

$$E(Y) = \int \cdots \int_{R^n} r(x_1, \ldots, x_n) f(x_1, \ldots, x_n)\, dx_1 \cdots dx_n,$$

if the mean exists. Similarly, if $X_1, \ldots, X_n$ have a discrete joint distribution with p.f. $f(x_1, \ldots, x_n)$, the mean of $Y = r(X_1, \ldots, X_n)$ is

$$E(Y) = \sum_{\text{All } x_1, \ldots, x_n} r(x_1, \ldots, x_n) f(x_1, \ldots, x_n),$$

if the mean exists. ■

Example 4.1.16 **Determining the Expectation of a Function of Two Variables.** Suppose that a point (X, Y) is chosen at random from the square S containing all points (x, y) such that $0 \le x \le 1$ and $0 \le y \le 1$. We shall determine the expected value of $X^2 + Y^2$.

Since X and Y have the uniform distribution over the square S, and since the area of S is 1, the joint p.d.f. of X and Y is

$$f(x, y) = \begin{cases} 1 & \text{for } (x, y) \in S, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$E(X^2 + Y^2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x^2 + y^2) f(x, y)\, dx\, dy$$
$$= \int_0^1 \int_0^1 (x^2 + y^2)\, dx\, dy = \frac{2}{3}.$$ ◀

Note: More General Distributions. In Example 3.2.7, we introduced a type of distribution that was neither discrete nor continuous. It is possible to define expectations for such distributions also. The definition is rather cumbersome, and we shall not pursue it here.

Summary

The expectation, expected value, or mean of a random variable is a summary of its distribution. If the probability distribution is thought of as a distribution of mass along the real line, then the mean is the center of mass. The mean of a function r of a random variable X can be calculated directly from the distribution of X without first finding the distribution of $r(X)$. Similarly, the mean of a function of a random vector $\boldsymbol{X}$ can be calculated directly from the distribution of $\boldsymbol{X}$.

Exercises

1. Suppose that X has the uniform distribution on the interval $[a, b]$. Find the mean of X.

2. If an integer between 1 and 100 is to be chosen at random, what is the expected value?

3. In a class of 50 students, the number of students n_i of each age i is shown in the following table:

Age i	n_i
18	20
19	22
20	4
21	3
25	1

If a student is to be selected at random from the class, what is the expected value of his age?

4. Suppose that one word is to be selected at random from the sentence THE GIRL PUT ON HER BEAUTIFUL RED HAT. If X denotes the number of letters in the word that is selected, what is the value of $E(X)$?

5. Suppose that one letter is to be selected at random from the 30 letters in the sentence given in Exercise 4. If Y denotes the number of letters in the word in which the selected letter appears, what is the value of $E(Y)$?

6. Suppose that a random variable X has a continuous distribution with the p.d.f. f given in Example 4.1.6. Find the expectation of $1/X$.

7. Suppose that a random variable X has the uniform distribution on the interval [0, 1]. Show that the expectation of $1/X$ is infinite.

8. Suppose that X and Y have a continuous joint distribution for which the joint p.d.f. is as follows:

$$f(x, y) = \begin{cases} 12y^2 & \text{for } 0 \le y \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the value of $E(XY)$.

9. Suppose that a point is chosen at random on a stick of unit length and that the stick is broken into two pieces at that point. Find the expected value of the length of the longer piece.

10. Suppose that a particle is released at the origin of the xy-plane and travels into the half-plane where $x > 0$. Suppose that the particle travels in a straight line and that the angle between the positive half of the x-axis and this line is α, which can be either positive or negative. Suppose, finally, that the angle α has the uniform distribution on the interval $[-\pi/2, \pi/2]$. Let Y be the ordinate of the point at which the particle hits the vertical line $x = 1$. Show that the distribution of Y is a Cauchy distribution.

11. Suppose that the random variables $X_1, \ldots, X_n$ form a random sample of size n from the uniform distribution on the interval [0, 1]. Let $Y_1 = \min\{X_1, \ldots, X_n\}$, and let $Y_n = \max\{X_1, \ldots, X_n\}$. Find $E(Y_1)$ and $E(Y_n)$.

12. Suppose that the random variables $X_1, \ldots, X_n$ form a random sample of size n from a continuous distribution for which the c.d.f. is F, and let the random variables Y_1 and Y_n be defined as in Exercise 11. Find $E[F(Y_1)]$ and $E[F(Y_n)]$.

13. A stock currently sells for \$110 per share. Let the price of the stock at the end of a one-year period be X, which will take one of the values \$100 or \$300. Suppose that you have the option to buy shares of this stock at \$150 per share at the end of that one-year period. Suppose that money

could earn 5.8% risk-free over that one-year period. Find the risk-neutral price for the option to buy one share.

14. Consider the situation of pricing a stock option as in Example 4.1.14. We want to prove that a price other than \$20.19 for the option to buy one share in one year for \$200 would be unfair in some way.

a. Suppose that an investor (who has several shares of the stock already) makes the following transactions. She buys three more shares of the stock at \$200 per share and sells four options for \$20.19 each. The investor must borrow the extra \$519.24 necessary to make these transactions at 4% for the year. At the end of the year, our investor might have to sell four shares for \$200 each to the person who bought the options. In any event, she sells enough stock to pay back the amount borrowed plus the 4 percent interest. Prove that the investor has the same net worth (within rounding error) at the end of the year as she would have had without making these transactions, no matter what happens to the stock price. (A combination of stocks and options that produces no change in net worth is called a *risk-free portfolio*.)

b. Consider the same transactions as in part (a), but this time suppose that the option price is \$$x$ where $x < 20.19$. Prove that our investor loses $|4.16x - 84|$ dollars of net worth no matter what happens to the stock price.

c. Consider the same transactions as in part (a), but this time suppose that the option price is \$$x$ where $x > 20.19$. Prove that our investor gains $4.16x - 84$ dollars of net worth no matter what happens to the stock price.

The situations in parts (b) and (c) are called *arbitrage opportunities*. Such opportunities rarely exist for any length of time in financial markets. Imagine what would happen if the three shares and four options were changed to three million shares and four million options.

15. In Example 4.1.14, we showed how to price an option to buy one share of a stock at a particular price at a particular time in the future. This type of option is called a *call option*. A *put option* is an option to sell a share of a stock at a particular price \$$y$ at a particular time in the future. (If you don't own any shares when you wish to exercise the option, you can always buy one at the market price and then sell it for \$$y$.) The same sort of reasoning as in Example 4.1.14 could be used to price a put option. Consider the same stock as in Example 4.1.14 whose price in one year is X with the same distribution as in the example and the same risk-free interest rate. Find the risk-neutral price for an option to sell one share of that stock in one year at a price of \$220.

16. Let Y be a discrete random variable whose p.f. is the function f in Example 4.1.4. Let $X = |Y|$. Prove that the distribution of X has the p.d.f. in Example 4.1.5.

4.2 Properties of Expectations

In this section, we present some results that simplify the calculation of expectations for some common functions of random variables.

Basic Theorems

Suppose that X is a random variable for which the expectation $E(X)$ exists. We shall present several results pertaining to the basic properties of expectations.

Theorem 4.2.1 Linear Function. If $Y = aX + b$, where a and b are finite constants, then

$$E(Y) = aE(X) + b.$$

Proof We first shall assume, for convenience, that X has a continuous distribution for which the p.d.f. is f. Then

$$\begin{aligned} E(Y) = E(aX + b) &= \int_{-\infty}^{\infty} (ax + b) f(x)\, dx \\ &= a \int_{-\infty}^{\infty} x f(x)\, dx + b \int_{-\infty}^{\infty} f(x)\, dx \\ &= aE(X) + b. \end{aligned}$$

A similar proof can be given for a discrete distribution. ■